TECHNO INDIA SILIGURI INSTITUTE OF TECHNOLOGY
A Satyam Roychowdhury Initiative

Introduction to Python Programming

Introduction to Python, Python Basics, Variables, Data Types, Type Conversion, Keywords, Operators, Conditional Statements (if, elif, else)

Introduction to Python

Python is a high-level, interpreted programming language known for:

- Simple and readable syntax
- Wide applications (AI, Data Science, Web Development, Automation)
- Large community support

```
>>> print("Hello Students")
Hello Students
```

Features of Python

- Easy to learn and use
- Platform-independent
- Open-source
- Supports multiple programming paradigms (procedural, OOP)

Variables

- A variable is used to store data.
- Must start with a letter or underscore
- Cannot start with a number
- Cannot use keywords
- Python automatically detects the type:

```
>>> x=12
>>> print(x)
12
>>> x="SIT"
>>> print(x)
SIT
```

Which of the variable names is correct for a Python variable?

A. if@

B. if

C. 1if

D. _if

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Data Type

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Basic Datatypes

Type	Example
int	12
float	12.65
string	“MCA”
bool	True

Collection Datatypes

Type	Example
list	[12,54,”hhkh”]
set	{1,5,89}
tuple	(12,34,”fght”,10)
dict	{“EE”:12,”ECE”:34,”CIVIL”:89}

Input/output

- The input() function is used to take user input from the keyboard.
- Syntax: $\text{var} = input(\text{"Input Message"})$
- The print() function displays output to the console. It can output strings, variables, and expressions.
- Syntax: $print(object(s), sep = ' ', end = '\text{\textbackslash n}', file = sys.stdout, flush = False)$

```
>>> x=input("enter your data")
enter your data23
>>> print(x)
23
```

Type Conversion

- Implicit conversion: Implicit type conversion in Python is an automatic process where the interpreter changes one data type to another during an operation to prevent data loss or errors.
- Explicit Conversion: a programmer manually changes the data type of a value using built-in functions.

```
>>> x=10
>>> x=x/3
>>> print(x)
3.3333333333333335
```

```
>>> print(int(10/3))
3
```

Keywords

- Python keywords are special reserved words with predefined meanings and functions that the Python interpreter uses to understand the structure and logic of a program.
- They cannot be used as names for variables, functions, classes, or any other identifiers.

- **Logic & Boolean:** `and`, `or`, `not`, `True`, `False`
- **Conditionals & Loops:** `if`, `elif`, `else`, `for`, `while`, `break`, `continue`
- **Functions & Classes:** `def`, `return`, `lambda`, `class`
- **Exceptions & Context:** `try`, `except`, `finally`, `raise`, `with`
- **Importing & Variables:** `import`, `from`, `as`, `global`, `nonlocal`
- **Memory & Structure:** `in`, `is`, `del`, `pass`, `None`

Which of the followings is output function of Python programming language?

A. write(“Hello Students”)

B. printf(“Hello Students”)

C. print(“Hello Students”)

D. printing(“Hello Students”)

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Operators

Operators	Example
Arithmetic Operators	$+, -, *, /, **, //$
Relational Operator	$<, \leq, >, \geq, ==, !=$
Logical Operators	and, or, not
Assignment Operators	$=, +=, -=, *=$

Conditional Statements

- Python's condition control statements, also known as conditional statements, allow a program to make decisions and execute different code blocks based on whether a condition evaluates to True or False.
- **if statement**: Executes a block of code only if the condition is True

```
amnt=int(input("enter purchase amount"))
discount=0
if amnt>=10000:
    discount=amnt*0.10 #10 percent discount on
print(f"Pay :{amnt-discount}")
```

```
enter purchase amount583
Pay :583
>>>
=== RESTART: C:/Users/Tur
enter purchase amount12000
Pay :10800.0
```

Conditional Statements

- if-else statement: Executes the if block if the condition is True, otherwise executes the else block.

```
amnt=int(input("enter purchase amount"))
discount=0
if amnt>=5000:
    discount=amnt*0.10 #10 percent discount on purch
else:
    discount=amnt*0.05
print(f"Pay :{amnt-discount}")
```

```
enter purchase amount2000
Pay :1900.0
>>>
=== RESTART: C:/Users/Tu
enter purchase amount6000
Pay :5400.0
```

Conditional Statements

- if-elif-else statement (ladder): Checks multiple conditions sequentially. The code block for the first True condition is executed, and subsequent conditions are skipped.

```
amnt=int(input("enter purchase amount"),
discount=0
if amnt<=5000:
    discount=amnt*0.05 #10 percent discount o
elif amnt<=10000.00:
    discount=amnt*0.10
else:
    discount=amnt*0.15
int(f"Pay :{amnt-discount}")
```

```
enter purchase amount20000
Pay :17000.0
>>>
=== RESTART: C:/Users/Tur
enter purchase amount6000
Pay :5400.0
```

Quiz